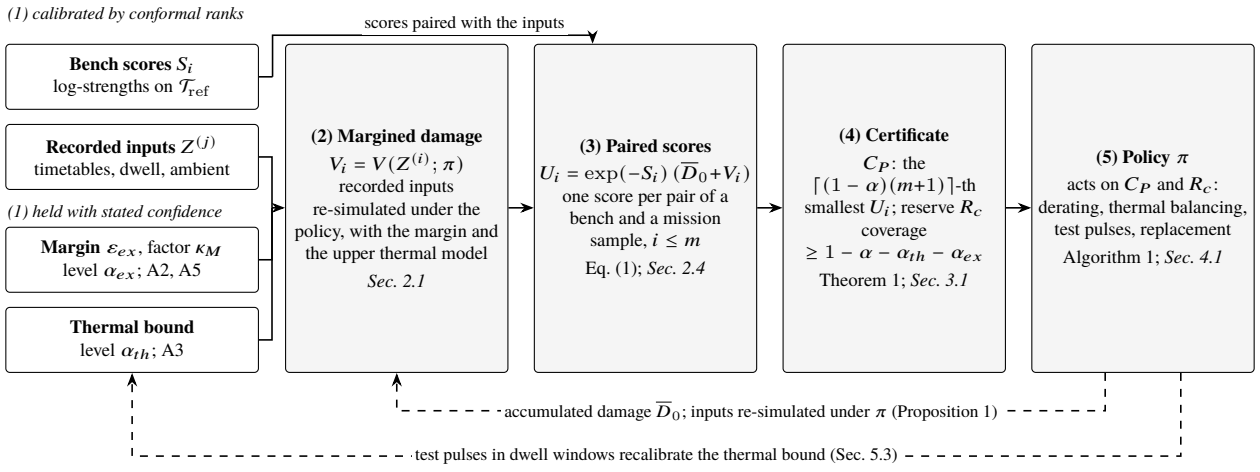
(1) calibrated by conformal ranks
scores paired with the inputs
Bench scores S_i
log-strengths on $\mathcal{T}_{\mathrm{ref}}$
Recorded inputs $Z^{(j)}$
timetables, dwell, ambient
(1) held with stated confidence
Margin ε_{ex}, factor κ_M
level α_{ex}; A2, A5
Thermal bound
level α_{th}; A3
(2) Margined damage
$V_i = V(Z^{(i)}; \pi)$
recorded inputs re-simulated under the policy, with the margin and the upper thermal model
Sec. 2.1
(3) Paired scores
$U_i = \exp(-S_i)\,(\overline{D}_0 + V_i)$
one score per pair of a bench and a mission sample, $i \le m$
Eq. (1); Sec. 2.4
(4) Certificate
C_P: the $\lceil (1-\alpha)(m+1) \rceil$-th smallest U_i; reserve R_c
coverage $\ge 1 - \alpha - \alpha_{th} - \alpha_{ex}$
Theorem 1; Sec. 3.1
(5) Policy π
acts on C_P and R_c: derating, thermal balancing, test pulses, replacement
Algorithm 1; Sec. 4.1
accumulated damage $\overline{D}_0$; inputs re-simulated under π (Proposition 1)
test pulses in dwell windows recalibrate the thermal bound (Sec. 5.3)